

# Chapter One

## Bohr Sommerfeld Theory of Quantization

### 1.1 Bohr Sommerfeld Theory of Quantization

For any physical system in which the coordinates are periodic function of time, there exist a quantum condition for each coordinate. These quantum conditions are

$$\oint p_q dq = n_q h$$

Where  $q$  is one of generalized coordinate  $p_q$  is the momentum associated with that coordinates,  $n_q$  is the quantum number which takes on integral values, and  $\oint$  means that integration is taken over one period of the coordinate  $q$ . The  $h$  is famous plank's constant which has value of  $6.6 \times 10^{-34}$  Joule-sec.

The diagram between  $q$  along horizontal axis and  $p_q$  along vertical axis are identified as phase diagram. In classical mechanics the area under the phase diagram is continuous but according to Bohr Sommerfeld theory it can be stated that area under the phase diagram is not continuous rather discrete in nature.

**Example:** A particle of mass  $m$  is confined in one dimensional box of width  $a$ . Consider particle is free inside box. Using Bohr Sommerfeld theory find the quantized energy of Particle.

**Solution:**

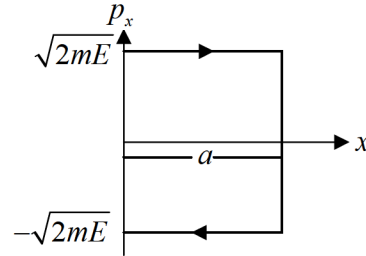
$$E = \frac{p_x^2}{2m}$$

$$\oint p_x dx = n_x h$$

$$\int_0^a \sqrt{2mE} dx + \int_a^0 (-\sqrt{2mE}) dx = n_x h, n_x = 1, 2, 3, \dots$$

$$2 \int_0^a \sqrt{2mE} dx = n_x h, n_x = 1, 2, 3, \dots \Rightarrow 2\sqrt{2mE}a = n_x h \Rightarrow E = \frac{n_x^2 h^2}{8ma^2} \Rightarrow \text{where } n_x = 1, 2, 3, \dots$$

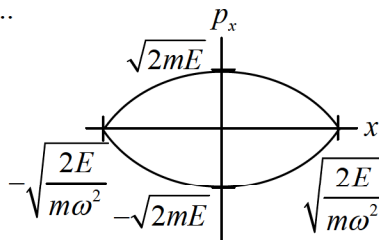
One can see energy is quantized as  $E_{n_x} = \frac{n_x^2 h^2}{8ma^2}$



**Example:** The classical energy of harmonic oscillator is given as  $E = \frac{p_x^2}{2m} + \frac{1}{2}m\omega^2 x^2$ . Using Bohr Sommerfeld theory find the possible quantized energy.

**Solution:**  $E = \frac{p_x^2}{2m} + \frac{1}{2}m\omega^2 x^2 \Rightarrow p_x = \sqrt{2m\left(E - \frac{1}{2}m\omega^2 x^2\right)}$

$\oint p_x dx = n_x h$  where  $n_x = 1, 2, 3, \dots$



$$4 \left(\frac{2E}{m\omega^2}\right)^{1/2} \int_0^{\left(\frac{2E}{m\omega^2}\right)^{1/2}} \sqrt{2m\left(E - \frac{m\omega^2 x^2}{2}\right)} dx = n_x h \Rightarrow \sqrt{2mE} \cdot 4 \int_0^{\left(\frac{2E}{m\omega^2}\right)^{1/2}} \sqrt{1 - \frac{m\omega^2 x^2}{2E}} dx = n_x h$$

Put the substitution  $\left(\frac{m\omega^2}{2E}\right)^{1/2} x = t \Rightarrow dx = \left(\frac{2E}{m\omega^2}\right)^{1/2} dt$  and

The limit of integration varied as  $(x=0, t=0)$  and  $\left(x=\left(\frac{2E}{m\omega^2}\right)^{1/2}, t=1\right)$

The integration can be evaluated as  $4\sqrt{2mE} \sqrt{\frac{2E}{m\omega^2}} \int_0^1 \sqrt{1-t^2} dt = n_x h, n_x = 1, 2, 3, \dots$

Again put  $t = \sin \theta$ ,  $dt = \cos \theta d\theta$  and limit will change from 0 to  $\frac{\pi}{2}$

$$4\sqrt{2mE} \sqrt{\frac{2E}{m\omega^2}} \cdot \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} \cos \theta d\theta = n_x h \Rightarrow 4\sqrt{2mE} \cdot \sqrt{\frac{2E}{m\omega^2}} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

where  $\int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \frac{\pi}{4}$

$$4\sqrt{2mE} \cdot \sqrt{\frac{2E}{m\omega^2}} \cdot \frac{\pi}{4} = n_x h \Rightarrow \frac{2\pi E}{\omega} = n_x h \Rightarrow E = n_x \frac{h\omega}{2\pi}$$

So quantized energy is  $E_{n_x} = n_x h\omega$  where  $n_x = 1, 2, 3, \dots$

**Example:** The classical energy of system is given as  $E = \frac{p_x^2}{2m} + kx^4$ .

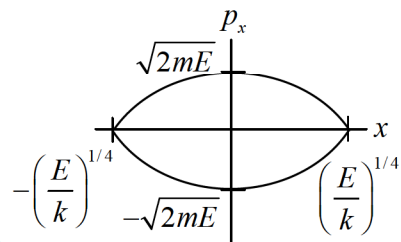
(a) For given value of Energy  $E$  area under the phase diagram is proportional to  $E^{3/4}$

(b) Using Bohr Sommerfeld theory the possible quantized energy proportional to  $(n_x)^\alpha$  where  $n_x$  is Bohr quantization number then find the value of  $\alpha$ .

**Solution:**  $E = \frac{p_x^2}{2m} + kx^4 \Rightarrow p_x = \sqrt{2m(E - kx^4)}$

Where area under the phase curve is  $\oint p_x dx = A$

$$A = 4 \int_0^{\left(\frac{E}{k}\right)^{1/4}} \sqrt{2m(E - kx^4)} dx = n_x h \Rightarrow \sqrt{2mE} \cdot 4 \int_0^{\left(\frac{E}{k}\right)^{1/4}} \sqrt{1 - \frac{kx^4}{E}} dx$$



Put the substitution  $\left(\frac{k}{E}\right)^{1/4} x = t \Rightarrow dx = \left(\frac{E}{k}\right)^{1/4} dt$  and

The limit of integration varied as  $(x=0, t=0)$  and  $\left(x=\left(\frac{E}{k}\right)^{1/4}, t=1\right)$

The integration can be evaluated as  $A = 4\sqrt{2mE} \left(\frac{E}{k}\right)^{1/4} \int_0^1 \sqrt{1-t^4} dt$

The integration  $\int_0^1 \sqrt{1-t^4} dt = \beta$  where  $\beta$  some constant value ( $0 < \beta < 1$ )

So  $A \propto \sqrt{2mE} \left(\frac{E}{k}\right)^{1/4} \Rightarrow A \propto E^{\left(\frac{1}{2} + \frac{1}{4}\right)} \Rightarrow A \propto E^{3/4}$

$\propto n_x \Rightarrow E \propto (n_x)^{4/3}$

(b) From Bohr Sommerfeld theory  $\oint p_x dx = n_x h$  where  $n_x = 1, 2, 3, \dots$

$$\oint p_x dx = A = n_x h \propto E^{3/4} \propto n_x \Rightarrow E \propto n_x^{4/3}$$

So quantized value of  $E_{n_x} \propto n_x^{4/3}$  so value of  $\alpha$  is  $\frac{4}{3}$

# Chapter Two

## Bohr Quantization Rule for Hydrogen Atom

### 1.2 Bohr Quantization Rule for Hydrogen Atom

Bohr give classical model of hydrogen atom in which electron moving in a circular orbit of radius  $r$  has an angular momentum  $mvr$  which is constant. The generalized coordinate for is assume to be  $\theta$  which vary from  $0 \leq \theta \leq 2\pi$  and conjugate momentum is  $p_\theta$  which is angular momentum  $L_z$  (we assume motion is confine into  $x-y$  plane so angular momentum is in  $z$  direction)

According to Bohr quantization rule

$$\oint p_\theta dq = n_q h \Rightarrow \oint p_\theta d\theta = nh \Rightarrow \oint L_z d\theta = nh \text{ where } n = 1, 2, 3 \dots$$

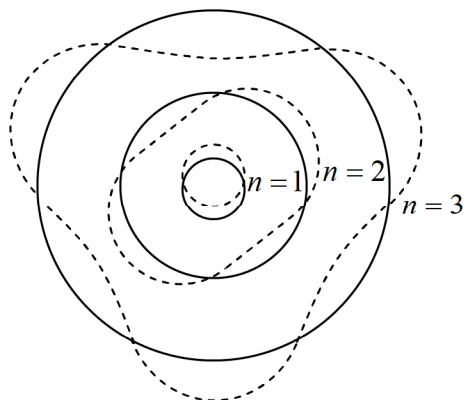
$$L_z \int_0^{2\pi} d\theta = nh \Rightarrow mvr = \frac{nh}{2\pi} \Rightarrow mvr = n\hbar \text{ which is very famous Bohr quantization rule .}$$

## De Broglie wavelength and hydrogen atom

From Bohr quantization rule  $mvr = n\hbar$

$$pr = n\hbar \text{ from de-Broglie hypothesis } p = \frac{h}{\lambda}$$

$$\frac{h}{\lambda} r = n\hbar = \frac{nh}{2\pi} \Rightarrow 2\pi r = n\lambda \text{ Where } n = 1, 2, 3 \dots$$



**Figure 7.5**

The allowed orbits are those in which the circumference of the orbit can contain exactly an integral number of de Broglie wavelengths. It can be imagined that electron is moving in a circular orbit at constant speed with the associated wave following the electron. The wave of wavelength  $\lambda$  is wrapped repeatedly around the circular orbit.

In the above figure de-Broglie standing wave interpretation satisfies Bohr quantization rule

## Bohr Model of Hydrogen Atom

In atomic physics, the **Bohr model**, introduced by Niels Bohr in 1913, depicts the atom as small, with a positively charged nucleus surrounded by electrons that travel in circular orbits around the nucleus—similar in structure to the solar system, but with attraction provided by electrostatic forces rather than gravity.

He suggested that electrons could only have certain **classical** motions:

1. Electrons in atoms orbit the nucleus which consists of single positive charge particle proton.

There is only external force action on electron is electrostatic force.

2. The electrons can only orbit stably, without radiating, in certain orbits (called by Bohr the “stationary orbits”): at a certain discrete set of distances from the nucleus. These orbits are associated with definite energies and are also called energy shells or energy levels. In these orbits, the electron’s acceleration does not result in radiation and energy loss as required by classical electromagnetism.

The significance of the Bohr model is that the laws of classical mechanics apply to the motion of the electron about the nucleus *only when restricted by a quantum rule*. The angular momentum  $L_z = mvr = n\hbar$  is restricted to be an integer multiple of a fixed unit:

$$L_z = n\hbar \text{ where } n = 1, 2, 3 \dots \text{ and}$$

$n$  is called the principle quantum number, and  $h = \frac{h}{2\pi}$ . The lowest value of  $n$  is 1. This gives a

smallest possible orbital radius of  $0.592A^0$  known as the **Bohr Radius**.

3. Electrons can only gain and lose energy by jumping from one allowed orbit to another, absorbing or emitting electromagnetic radiation with a frequency  $\nu$  determined by the energy difference of the levels according to the *Planck relation*:

$$\Delta E = E_f - E_i = h\nu$$

where  $h$  is Planck's constant. The frequency of the radiation emitted at an orbit of period  $T$  is as it would be in classical mechanics; it is the reciprocal of the classical orbit period:  $\nu = \frac{1}{T}$

## Hydrogen Atom

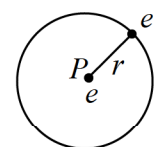
A **hydrogen atom** is an atom of the chemical element hydrogen. The electrically neutral atom contains a single positively charged proton and a single negatively charged electron bound to the nucleus by the Coulomb force. According to Bohr electron revolve about the nucleus in different quantized circular orbits whose angular momentum is given by  $L = n\hbar$  where  $n = 1, 2, 3, \dots$ .

The electron is held in a circular orbit by electrostatic attraction. The centripetal force is equal

to the Coulomb force.  $\frac{m_e v^2}{r} = \frac{k_e e^2}{r^2}$

where  $m_e$  and  $e$  the electron's mass and charge of the electron,

$k_e = \frac{1}{4\pi\epsilon_0}$  is identified as Coulomb's constant.



If  $v$  is velocity of electrons in orbit, the electron's speed at any radius:

$$v = \sqrt{\frac{k_e e^2}{m_e r}} \Rightarrow v = \sqrt{\frac{e^2}{m_e 4\pi\epsilon_0 r}}$$

The electron's total energy at any radius:  $E = T + V = \frac{m_e v^2}{2} - \frac{k_e e^2}{r}$

Putting the value of  $v$  one will get  $E = -\frac{k_e e^2}{2r}$ . The total energy is half the potential energy.

From the quantization the angular momentum  $L_z = n\hbar = m_e v r = n\hbar$

Substituting the expression for the velocity gives an equation for  $r$  in terms of  $n$ :

$$\sqrt{k_e e^2 m_e r} = n\hbar \text{ allowed orbit radius at any } n \text{ is: } r_n = \frac{n^2 \hbar^2}{k_e e^2 m_e} \Rightarrow r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e e^2}$$

For  $n=1$   $r_1 = a_0 = 0.53 \times 10^{-10} m$  called the Bohr radius.

Hence radius of hydrogen atom is quantized one can also see speed of electron is also quantized

$$v = \sqrt{\frac{e^2}{m_e 4\pi\epsilon_0 r}} \Rightarrow \sqrt{\frac{e^4}{(4\pi\epsilon_0)^2 n^2 \hbar^2}} = \frac{e^2}{4\pi\epsilon_0 n \hbar} = \frac{2.2 \times 10^6 m/sec}{n}$$

The energy of the  $n^{\text{th}}$  level for any atom is determined by the radius and quantum number:

$$E_n = -\frac{k_e e^2}{2r_n} = -\frac{(k_e e^2)^2 m_e}{2\hbar^2 n^2} = -\frac{m_e}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = \frac{-13.6}{n^2} eV$$

The Rydberg energy  $R_E$  which is given by  $R_E = \frac{(k_e e^2)^2 m_e}{2\hbar^2} \Rightarrow \frac{m_e}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2$

## 1.3 Hydrogen Like Atom

A **hydrogen-like atom/ion** (usually called a "hydrogenic atom") is any atomic nucleus bound to one electron and thus is is electronic with hydrogen. These atoms or ions can carry the positive charge where is the atomic number of the atom. Examples of hydrogen-like atoms/ions are Hydrogen itself,  $He^+$ ,  $Li^{2+}$ ,  $Be^{3+}$  and  $B^{4+}$ . Because hydrogen-like atoms/ions are two-particle systems with an interaction depending only on the distance between the two particles.

### **A correction in radius velocity and energy due to atomic number $Z$ of atom**

If the atomic number of hydrogen like atom is  $Z$  then potential energy of system will

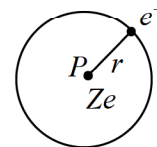
$$V(r) = -\frac{q_1 q_2}{4\pi\epsilon_0 r^2} = -\frac{Ze.e}{4\pi\epsilon_0 r^2}. \text{ Accordingly there is correction in physical measurement of}$$

hydrogen like Atom.

$$1 \text{ Correction in radius } r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e Ze.e} = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e Ze^2} = \frac{0.53 n^2}{Z} A^0$$

$$2 \text{ Correction in speed } v_n = \frac{Ze.e}{4\pi\epsilon_0 n \hbar} = \frac{Z}{n} \times 2.2 \times 10^6 m/sec$$

$$3 \text{ Correction in energy } E_n = -\frac{m_e}{2\hbar^2} \left( \frac{Ze.e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -Z^2 \frac{13.6}{n^2}$$



### **Correction Due to Finite Nuclear Mass**

In Physical measurement of hydrogen atom it is assumed that mass of nucleus is infinite in comparison of electron. But if mass of nucleus is finite then the system is two particle system in

which the orbiting mass is replaced by reduce mass  $\mu = \frac{m_n \cdot m_e}{m_n + m_e}$  where  $m_n$  is mass of nucleus and  $m_e$  mass of electron. (The derivation is in equivalent to two body planetary motion discussed in mechanics)

$$1. \text{ Correction in radius } r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{\mu Z e^2} = \frac{m_e 4\pi\epsilon_0 n^2 \hbar^2}{\mu m_e Z e^2} = 0.53 \frac{m_e}{\mu} \frac{n^2}{Z} A^0$$

$$2. \text{ Correction in speed } v_n = \frac{Z e e}{4\pi\epsilon_0 n \hbar} = \frac{Z}{n} \times 2.2 \times 10^6 \text{ m/sec no Correction due to reduce mass}$$

$$3. \text{ Correction in energy } E_n = -\frac{\mu}{2\hbar^2} \left( \frac{Z e e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -\frac{\mu}{m_e} \frac{m_e}{2\hbar^2} \left( \frac{Z e e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -Z^2 \frac{\mu}{m_e} \frac{13.6}{n^2}$$

**Relationship between kinetic energy  $T$  potential energy  $U$  and total energy  $E$**

$$U = -\frac{k_e e^2}{r}, T = \frac{k_e e^2}{2r}, E = -\frac{k_e e^2}{2r}$$

$$U = \frac{E}{2}, T = -E$$

### The spectrum of hydrogen atom

The electron may undergo a transition to some other stationary state either by absorbing energy and moving up to a higher energy state or giving off energy and moving down. The transition between energy levels are constantly occurring. The atom absorbed or released energy in form of electromagnetic radiation.

Energy of radiation corresponds to difference of energy between transition levels initially in  $n_i$  to final level  $n_f$

$$\text{So } E_\gamma = E_f - E_i = \left( \frac{-13.6}{n_f^2} \right) - \left( \frac{-13.6}{n_i^2} \right) = 13.6 \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

If  $E_\gamma$  is positive means  $n_f > n_i$  mean hydrogen atom absorb radiation

If  $E_\gamma$  is positive means  $n_f < n_i$  mean hydrogen atom release radiation

### The wavelength of radiation and Rydberg constant

According to Planck formula the energy of radiation is given by  $E_\gamma = h\nu = \frac{hc}{\lambda}$

$$E_\gamma = |E_f - E_i| = \frac{m_e}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right|$$

$$\frac{hc}{\lambda} = \frac{2\pi\hbar c}{\lambda} = \frac{m_e}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right| \Rightarrow \frac{1}{\lambda} = \frac{m_e}{4\pi c \hbar^3} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right|$$

$$\frac{1}{\lambda} = R_\infty \left| \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right| \text{ where } R_\infty = \frac{m_e}{4\pi c \hbar^3} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 = 1.097 \times 10^7 \text{ m}^{-1}$$

known as Rydberg constant  $R_\infty$

### Correction in Rydberg constant for hydrogen like atom

If  $Z$  is atomic number and  $\mu$  is the reduce mass of hydrogen like atom the

$$R_\mu = \frac{\mu Z^2}{4\pi c \hbar^3} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 = \frac{\mu}{m_e} \cdot Z^2 \times R_\infty \Rightarrow R_\mu = \frac{\mu}{m_e} \cdot Z^2 \times 1.097 \times 10^7 \text{ m}^{-1}$$

### Spectral Series

When an electric discharge is passed through gaseous hydrogen, the  $H_2$  molecules dissociate and the energetically excited  $H$  atoms that are produced emit light of discrete frequencies, producing a spectrum of a series of 'lines'.

(i) **Lyman Series:**  $\frac{1}{\lambda} = R \left( \frac{1}{1^2} - \frac{1}{n^2} \right)$  ;

$n = 2, 3, 4 \dots$  (In U.V. region), where,  $R$  is Rydberg constant ( $1.097 \times 10^7 \text{ m}^{-1}$ )

(ii) **Balmer Series:**

$$\frac{1}{\lambda} = R \left( \frac{1}{2^2} - \frac{1}{n^2} \right) \quad n = 3, 4, 5 \dots \text{ (in visible region), } n = 3 \text{ for } H_\alpha \text{ line } n = 4 \text{ } H_\beta \text{ line}$$

$n = 5 \text{ } H_\gamma \text{ line}$

(iii) **Paschen Series:**

$$\frac{1}{\lambda} = R \left( \frac{1}{3^2} - \frac{1}{n^2} \right) \quad n = 4, 5, 6 \dots \text{ (Near Infra Red)}$$

(iv) **Bracket Series:**

$$\frac{1}{\lambda} = R \left( \frac{1}{4^2} - \frac{1}{n^2} \right); n = 5, 6, 7 \dots \text{ (Infra Red)}$$

(v) **Pfund Series:**  $\frac{1}{\lambda} = R \left( \frac{1}{5^2} - \frac{1}{n^2} \right); n = 6, 7, 8 \dots \text{ (Far Infra Red)}$

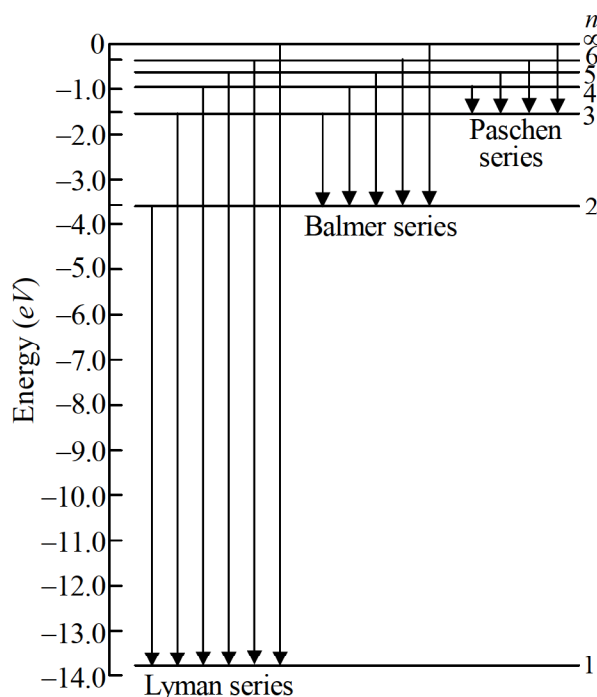


Figure 7.6

**Ionization energy of hydrogen atom:** If electron is in ground state  $n=1$ , then Minimum energy required by hydrogen atom such that electron became unbounded. Electron became unbounded it will be in continuum state which has energy  $E \geq 0$

Initial state is  $n=1$ ,  $E_1 = -\frac{13.6}{1^2} = -13.6\text{eV}$  the final state is  $n = \infty$   $E_f = -\frac{13.6}{\infty} = 0$

$\Delta E$  is Energy required by atom such that electron will transit from  $E_1 \rightarrow E_f$

$$\Delta E = E_f - E_1 = 0 - (-13.6\text{eV}) = 13.6\text{eV}$$

**Example:** If hydrogen atom at rest it emits a photon in going from  $n=5$  state to  $n=1$  state is  $m/s$ . Find the recoil speed of a hydrogen atom in meter/sec after the transition

**Solution:** If Momentum of recoil hydrogen atom  $p_H = mv$  where mass of hydrogen atom  $m = 1.67 \times 10^{-27} \text{ kg}$

Momentum of emitted photon  $P_\gamma = \frac{\Delta E}{c}$

Linear momentum is conserved in the recoil process so  $P_H = P_\gamma$

$$\Delta E = E_5 - E_1 = -13.6 \left[ \frac{1}{5^2} - \frac{1}{1^2} \right] \text{eV} = (13.6) \left( \frac{24}{25} \right) \text{eV} = \frac{13.6 \times 24}{25} \times (1.6 \times 10^{-19}) \text{ J}$$

$$\Rightarrow \Delta E = 2.1 \times 10^{-18} \text{ J}$$

$$P_H = P_\gamma \Rightarrow mv = \frac{\Delta E}{c} \quad \text{or} \quad v = \frac{\Delta E}{mc} \Rightarrow v = \frac{2.1 \times 10^{-18}}{(1.67 \times 10^{-27}) \times (3 \times 10^8)} = 4.19 \text{ m/s}$$

**Example:** A single electron orbits around a stationary nucleus of charge  $+Ze$ , where  $Z$  is a constant and  $e$  is the magnitude of the electronic charge. It requires  $47.2\text{eV}$  to excite the electron from the second Bohr orbit to the third Bohr orbit. Find

- (a) The value of  $Z$ .
- (b) The energy required to excite the electron from the third to the fourth Bohr orbit
- (c) The wavelength Of the electromagnetic radiation required to remove the electron from the first Bohr orbit to infinity.
- (d) The kinetic energy, potential energy, and the angular momentum of the electron in the first Bohr orbit.
- (e) The radius of the first Bohr orbit.

(The ionization energy of hydrogen atom  $= 13.6\text{eV}$ , Bohr radius  $= 5.3 \times 10^{-11}$  meter, velocity of light  $= 3 \times 10^8 \text{ m/sec}$ . Planck's constant  $= 6.6 \times 10^{-34} \text{ Joule - sec}$ )

**Solution:** (a) According to Bohr model, for hydrogen like atom  $E_n = -\frac{13.6Z^2}{n^2} eV$

$$\therefore (E_3 - E_2) = -13.6Z^2 \left[ \frac{1}{9} - \frac{1}{4} \right] \text{ or } 47.2 = +\frac{13.6Z^2 \times 5}{36} \Rightarrow Z^2 = \frac{47.2 \times 36}{13.6 \times 5} = 25 \text{ or } Z = 5$$

$$(b) E_4 - E_3 = -13.6Z^2 \left[ \frac{1}{4^2} - \frac{1}{3^2} \right] = \frac{13.6Z^2 \times 7}{9 \times 16} = \frac{13.6 \times 25 \times 7}{9 \times 16} = 16.53 eV$$

$$(c) \lambda \text{ for removing electron from } n=1 \text{ to } n=\infty \therefore E_\infty = 0 \quad E = -E_1 = +\frac{13.6}{(1)^2} \times Z^2 eV$$

$$E = 13.6 \times 25 \times 1.6 \times 10^{-19} J \Rightarrow \frac{hc}{\lambda} = 13.6 \times 25 \times 1.6 \times 10^{-19}$$

$$\Rightarrow \lambda = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{13.6 \times 25 \times 1.6 \times 10^{-19}} \Rightarrow \lambda = 3.64 \times 10^{-9} m$$

$$(d) \text{ Total energy in first orbit} = -\frac{13.6Z^2}{n^2}; \quad E = -\frac{13.6 \times 25}{1} = -340 eV$$

$$E = -340 eV \quad T = -E \text{ or } T = 340 eV$$

$$U = 2E = 2(-340) eV \Rightarrow U = -680 eV, \quad L_z = \frac{nh}{2\pi} = \frac{1 \times 1.66 \times 10^{-34}}{2\pi} = 1.05 \times 10^{-34} Js$$

$$(e) r_1 = \frac{\text{Bohr radius}}{Z} \quad \text{or } \Rightarrow r_1 = \frac{5.3 \times 10^{-11}}{5} \Rightarrow r_1 = 1.06 \times 10^{-11} m$$

**Example:** A double ionized lithium atom is hydrogen-like with atomic number 3.

(a) Find the wavelength of the radiation required to excite the electron in Li from the first to the third Bohr orbit. (Ionization energy of the hydrogen atom equal  $13.6 eV$ ).

(b) How many spectral lines are observed in the emission spectrum of the above excited system?

**Solution:** (i) For hydrogen like atom  $E_n = \left( -\frac{13.6}{n^2} Z^2 \right) eV$   $\lambda$  of radiation excite electron from  $n=1$  to  $n=3$ :

$$\text{Given: } Z = 3, \quad E_1 = -\frac{13.6}{(1)^2} \times (3)^2 = -13.6 \times 9 eV \text{ and } E_3 = -\frac{13.6}{(3)^2} \times (3)^2 = -13.6 eV$$

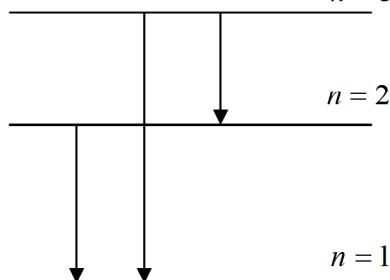
$$\therefore E_3 - E_1 = -13.6 - (-13.6 \times 9) \quad \Delta E = 13.6 \times 8$$

$$\text{or } \Delta E = 108.8 eV = 108.8 \times 1.6 \times 10^{-19} J \Rightarrow \frac{hc}{\lambda} = (108.8 \times 1.6 \times 10^{-19})$$

$$\lambda = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{108.8 \times 1.6 \times 10^{-19}} \Rightarrow \lambda = 1.137 \times 10^{-8} m = 113.7 \text{ \AA}$$

(b) Number of spectral lines in emission spectrum:

or Number of spectral lines =  $\frac{n(n-1)}{2}$       Number of spectral lines =  $\frac{3 \times 2}{2}$   
 $n = 3$



$\therefore$  Number of spectral lines = 3. The transition are  $3 \rightarrow 1, 3 \rightarrow 2, 2 \rightarrow 1$

**Example:** A muonic atom contains a nucleus of charge  $Ze$  and a negative muon,  $\mu^-$ , moving about it. The  $\mu^-$  is an elementary particle with charge  $-e$  and a mass that is 207 times as large as an electron mass.

- Calculate the muon-nucleus separation,  $D$ , of the first Bohr orbit of a muonic atom with  $Z = 1$ .
- Calculate the binding energy of a muonic atom with  $Z = 1$
- What is the wavelength of the first line in the Lyman series for such an atom?

**Solution:** (a) The reduced mass of the system, with  $m_{\mu^-} = 207m_e$  and  $M = 1836m_e$ , is, from (4-20)

$$\mu = \frac{207m_e \times 1836m_e}{207m_e + 1836m_e} = 186m_e$$

Then, from (4-16), with  $n = 1, Z = 1$  and  $m = 186m_e$ , we obtain

$$D_1 = \frac{4\pi\epsilon_0 \hbar^2}{186m_e e^2} = \frac{1}{186} \times 5.3 \times 10^{-11} m = 2.8 \times 10^{-13} m = 2.8 \times 10^{-3} \text{ \AA}$$

Therefore the  $\mu^-$  is much closer to the nuclear (proton) surface than is the electron in a hydrogen atom. It is this feature which makes such muonic atoms interesting, information about nuclear properties being revealed from their study.

(b) From (4-18) with  $Z = 1, n = 1$  and  $m = \mu = 186m_e$ , we have

$$E = -186 \frac{m_e e^4}{(4\pi\epsilon_0)^2} = -186 \times 13.6 eV = -2530 eV$$

as the ground state energy. Hence, the binding energy is  $2530 eV$ .

(c) From (4-22), with  $Z = 1$ , we have

$$\kappa = R_M \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

For the first Lyman line,  $n_i = 2$  and  $n_f = 1$ . In this case,  $R_M = (\mu / m_e) R_\infty = 186 R_\infty$ . Hence

$$\kappa = \frac{1}{\lambda} = 186 R_\infty \left( 1 - \frac{1}{4} \right) = 139.5 R_\infty$$

With  $R_\infty = 109737 \text{ cm}^{-1}$   $\lambda \approx 6.5 \text{ \AA}$

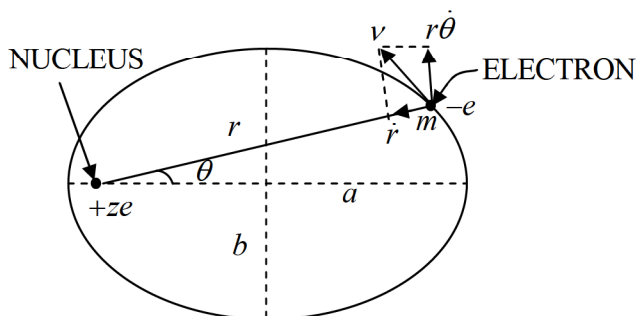
so that the Lyman lines lie in the  $x$  - ray part of the spectrum.  $X$  - ray techniques are necessary, therefore, to study the spectrum of muonic atoms.

# Chapter Three

## Sommerfeld extension of Bohr model

### 1.4 Sommerfeld extension of Bohr model

In Bohr model of hydrogen atom, the electron could move only in circular orbits. But Sommerfeld produce a advance model in which the electron revolved about proton in elliptical orbit where proton is at one of foci of ellipse. The model is exactly analogical to Kepler's model of sun and earth system.



From Sommerfeld quantization rule for radial direction  $\oint p_r dr = n_r h$

From Sommerfeld quantization rule for azimuthal direction  $\oint p_\theta d\theta = n_\theta h$

So  $p_\theta = \frac{n_\theta h}{2\pi}$

We know that radial momentum  $p_r = m\dot{r}$  and angular momentum  $p_\theta = mr^2\dot{\theta}$

So polar equation of ellipse is given by  $\frac{l}{r} = 1 + \varepsilon \cos \theta$  where  $l$  is latus rectum and  $\varepsilon$  is eccentricity.

Differentiating both sides, we will get

$$-\frac{l}{r^2} \dot{r} = -\varepsilon \sin \theta \dot{\theta} \Rightarrow \frac{l}{r^2} \dot{r} = \varepsilon \sin \theta \frac{p_\theta}{mr^2} \Rightarrow \dot{r} = \varepsilon \sin \theta \frac{p_\theta}{m} \Rightarrow p_r = m\dot{r} = p_\theta \varepsilon \sin \theta$$

$$-\frac{l}{r^2} \dot{r} = -\varepsilon \sin \theta \dot{\theta} \Rightarrow \frac{l}{r^2} \frac{dr}{dt} = \varepsilon \sin \theta \frac{d\theta}{dt} \Rightarrow dr = \frac{\varepsilon \sin \theta d\theta}{(1 + \varepsilon \cos \theta)^2}$$

$$\oint p_r dr = p_\theta \int_0^{2\pi} \varepsilon^2 \frac{\sin^2 \theta d\theta}{(1 + \varepsilon \cos \theta)^2} = n_r h$$

$$2\pi p_\theta \left( \frac{1}{\sqrt{1 - \varepsilon^2}} - 1 \right) = n_r h \Rightarrow 2\pi \frac{n_\theta h}{2\pi} \left( \frac{1}{\sqrt{1 - \varepsilon^2}} - 1 \right) = n_r h$$

$$\sqrt{1 - \varepsilon^2} = \frac{b}{a} = \frac{n_\theta}{n_r + n_\theta} \Rightarrow \frac{n_\theta}{n} = \frac{b}{a} \text{ where } n_\theta = 1, 2, 3 \dots, n_r = 0, 1, 2, 3 \dots \text{ and } n = 1, 2, 3 \dots$$

(If  $n_\theta = 0$  then  $b = 0$  which is not support the basic property of ellipse so  $n_\theta \neq 0$ )

Sommerfeld also calculated the other parameter of Hydrogen atom

## Energy

$$F = \frac{mz^2 e^4 \left( \frac{b}{a} \right)^2}{32\pi^2 \epsilon_0^2 p_\theta^2} \quad \frac{b}{a} = \frac{n_\theta}{n} \text{ and } p_\theta = \frac{kh}{2\pi} \text{ so } F = -\frac{mz^2 e^4}{8\epsilon_0^2 h^2} \left( \frac{1}{n^2} \right) \Rightarrow -\frac{mz^2}{2h^2} \left( \frac{e^2}{4\pi \epsilon_0} \right)^2$$

Size and shape of Sommerfeld's Elliptical orbits

$$\text{Semimajor Axis } a = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2} = \frac{4\pi \epsilon_0}{m z e^2} n^2 \Rightarrow \frac{n^2}{z} a_0$$

$$\text{Semi minor axis } b = \frac{nn_\theta}{z} a_0$$

Possible shape of orbit

$n = 1, n_\theta = 1, a = a_0, b = a_0$  which is circular in nature with radius  $a_0$

$n = 2, n_\theta = 2, a = 4a_0, b = 4a_0$  which is circular in nature with radius  $4a_0$

$n = 2, n_\theta = 1, a = 4a_0, b = 2a_0$  which is elliptical in nature.

## 1.5 Summary of solution Schrödinger wave equation for hydrogen atom with (this topic is detail discuss in quantum mechanics course)

The hydrogen atom consist of heavy essentially motionless proton (which can be put it at origin )of charge  $e$  together with a much lighter electron ( charge  $-e$ ) that orbits around it ,bound by mutual attraction from coulomb potential as  $V(r) = \frac{-e^2}{4\pi \epsilon_0 r}$  .the coulomb

potential is central potential so angular momentum of electron is conserve .so angular momentum operator  $\hat{L}^2$  will commute with Hamiltonian .  $[H, L^2] = 0$  If  $\psi(r, \theta, \phi)$  is solution of Schrodinger wave equation then  $\psi(r, \theta, \phi) = R(r) Y_l^m(\theta, \phi)$

Where  $R(r)$  is radial wave function and  $Y_l^m(\theta, \phi)$  is spherical harmonics.

The Schrodinger equation reduce to

$$\frac{-\hbar^2}{2m} \left[ \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right] + \left[ \frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right] R(r) = ER(r) \text{ where } V(r) = \frac{-e^2}{4\pi \epsilon_0 r}$$

Putting  $R(r) = \frac{u(r)}{r}$  the above equation reduces

$$\frac{-\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[ \frac{l(l+1)\hbar^2}{2mr^2} - \frac{e^2}{4\pi \epsilon_0 r} \right] u(r) = Eu(r) \Rightarrow \frac{-\hbar^2}{2m} \frac{d^2 u}{dr^2} = \left[ E - \frac{l(l+1)\hbar^2}{2mr^2} + \frac{e^2}{4\pi \epsilon_0 r} \right] u(r)$$

The general solution is given

$$\psi(r, \theta, \phi) = L_n^l(r) Y_l^m(\theta, \phi) \exp\left(-\frac{r}{na}\right) \text{ Where } L_n(r) \text{ is Laguerre Polynomial}$$

Where  $n = 1, 2, 3, \dots$  and  $l = 0, 1, 2, \dots, n-1$  can be found from  $j_{\max} + l + 1 = n$  if  $j_{\max} = 1$

$m = -l, l+1, -l+2, \dots, 0, 1, 2, \dots, l-1, l$  then  $l = n-2$

Where  $L_n(r)$  is Laguerre Polynomial

$$L_n(r) = \exp(r) \left( \frac{d}{dx} \right)^n (\exp(-r) r^n) \quad L_{n-l}^l(r) = (-1)^l \left( \frac{d}{dr} \right)^l L_n(r)$$

The general solution is given by  $\psi_{n,l,m} = \sqrt{\left( \frac{2}{na} \right)^3 \frac{(n-l-1)!}{2n(n+1)!}} \exp\left(-\frac{r}{na}\right) \left[ L_{n-l-1}^{2l+1} \left( \frac{2r}{na} \right) \right] Y_l^m(\theta, \phi)$

## The first few radial wave functions of hydrogen atom

$$R_{10} = 2a^{-3/2} \exp\left(-\frac{r}{a}\right) \text{ for } n=1 \text{ and } l=0$$

$$R_{20} = \frac{1}{\sqrt{2}} a^{-3/2} \left(1 - \frac{1}{2} \frac{r}{a}\right) \exp\left(-\frac{r}{2a}\right)$$

$$R_{21} = \frac{1}{\sqrt{24}} a^{-3/2} \frac{r}{a} \exp\left(-\frac{r}{2a}\right)$$

Ortho normal relation is given by  $\int_0^{\infty} R_n^* R_n r^2 dr = \delta_{n'n}$

## Some important spherical harmonics

$$|0,0\rangle = Y_0^0 = \frac{1}{\sqrt{4\pi}}, |1,0\rangle = Y_1^0 = \sqrt{\frac{1}{8\pi}} \sin \theta e^{i\phi}, |1,-1\rangle = Y_1^{-1} = \sqrt{\frac{3}{8}} \sin \theta e^{-i\phi}$$

Orthonormal relation of  $Y_l^m(\theta, \phi)$   $\int_0^{2\pi} \int_0^{\pi} Y_l^{m'}(\theta, \phi) Y_l^m(\theta, \phi) \sin \theta d\theta d\phi = \delta_{l'l} \delta_{m'm}$

which can be represented as  $\langle l', m' | l, m \rangle = \delta_{l'l} \delta_{m'm}$

## Orthonormal relation for total wave function

$\psi_{n,l,m}(r, \theta, \phi)$  where  $n$  is principle quantum number,  $l$  is orbital quantum number and  $m$  is magnetic quantum number.

$$\int_0^{\infty} \int_0^{\pi} \int_0^{2\pi} \psi_{n',l',m'}^* \psi_{n,l,m} r^2 \sin \theta dr d\theta d\phi = \delta_{n',n} \delta_{l',l} \delta_{m',m}$$

Effective potential, ground state energy and ionization energy

Effective potential is given by  $V_{\text{effective}} = \frac{\hat{L}^2}{2mr^2} - \frac{e^2}{4\pi\epsilon_0 r}$  is shown in fig.

**The bound state energy** is negative which is given by  $E_n = -\frac{m}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2}$

Ground state energy is minimum energy of electron in bound state which is given by

$$E_1 = -\frac{m}{2\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0} \right) = -13.6 \text{ eV}.$$

Ionization energy is the minimum energy that an electron in a hydrogen atom or has to absorb to come out of the influence of the electrostatic potential. If  $E_s$  is ionization energy is

$$E_s = E_f - E_i. \text{ Where } E_f \text{ is minimum final energy when electron will unbounded i.e. } E_f = 0. E_i$$

$$\text{is minimum initial energy when it is bounded i.e. } E_i = -13.6 \text{ eV} \quad E_s = 0 - (-13.6 \text{ eV}) = 13.6 \text{ eV}.$$